

Prediction of Self-induced Mechanoreceptive Sensor Readings in An Insect-inspired Active Tactile Sensing System

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1 Introduction

Many animals use actively moveable tactile sensors to explore the near-range space. While mammals like rats and cats use active whisker movements to detect and scan objects in the vicinity of the body, insects and crustaceans use antennae (or feelers) for obstacle localization, orientation and pattern recognition [1]. Compared to vision-based sensors, the tactile sense is independent of light conditions, it works at day and night. Recently, a number of artificial tactile sensing systems have been developed that were inspired by insect antennae or mammal whiskers. Insect antennae are highly sensorized and multimodal limbs of the head. As actively moving sensory organs, they are subject to self-stimulation, particularly of their mechanoreceptors. For an animal to adapt its behaviour to environmental conditions, it is important to identify external cues, e.g., true touch events, from self-stimulation during self-motion.

Here we use a bionic tactile sensor inspired by the antenna of the stick insect, *Carausius morosus*. It consists of an acceleration sensor mounted to the tip of an otherwise unsensorized probe, and is capable of near-range tactile localization and material classification [2]. However, if the tactile system is attached to a moving body, with each sensor being moved relative to the body, the mechanosensory signal will be corrupted by self-induced motion artefacts. Self-stimulation is a general problem of sensory systems on moving bodies, and of moving mechanoreceptors in particular. In [2], state-dependent modulation of input signals was used to suppress the detrimental effects of robot velocity on the antennal dynamics. Rather than modulating the input signal, [3] applied a forward model based on Echo State Networks (ESN) for predicting the expected sensory reading. The goal was to tell true contact events from false positives as the robot was moving along a path with bumps. Here, we expand this system to two antennae with 2 degrees of freedoms each, executing a complex sampling pattern in three dimensional (3D) space. Additionally, we have tested the prediction system in different scenarios (see section 2.1) which included not only different acceleration profiles of the robot, but also rhythmic perturbation by a spiked wheel, simulating rhythmic impacts of the feet of a legged robot.

In physiological motor control, a forward model is an internal approximation of the motor system that uses the current state of the system and the motor command (*efference copy*) to predict the next state [4]. The state affects the processing of sensory signals. Therefore, the forward model can be used to estimate the sensory consequences of a motor command. Self-induced movements can produce sensory inputs (*reafference*) that are indistinguishable from inputs caused externally (*afference*). A forward model can be used to cancel the reafferent signal (*corollary discharge*), and thus allow the external signals to be recovered [4, 5].

The forward model described here is based on an Echo State Network (ESN), a special type of artificial neural network. It has been shown that ESNs are suitable for chaotic time series predictions with a comparatively high accuracy [6]. In general, ESNs have a large, fixed, randomly initialized reservoir of recurrently and sparsely connected neurons. The output of the ESN is read out through a *linear* layer and the training of the output weight matrix is a linear regression task which is straightforward and fast compared to classical, gradient-based training methods. Here, inputs to the ESN are the motor commands which set the positions of the antennae, and a local proprioceptive signal which measures the vibrations of the robot platform.

The next section briefly describes the experimental setup, our approach, and the test scenarios. For more details of the experimental setup, see [3]. In section 3, the results are shown and discussed briefly. Then a summary of findings and contributions of our investigation are given. Finally, open questions will be discussed.

2 Approach

The data for training the ESN were obtained from several trials carried out with the moving robot (Erratic base, ERA-Videre Design LLC). The two antennae were mounted onto two pan-tilt-units, each allowing active rotation around two orthogonal axes, controlled via *servo* motors (HiTEC HS-755MG, see Figure 1a). Each antenna carried a 2-axes

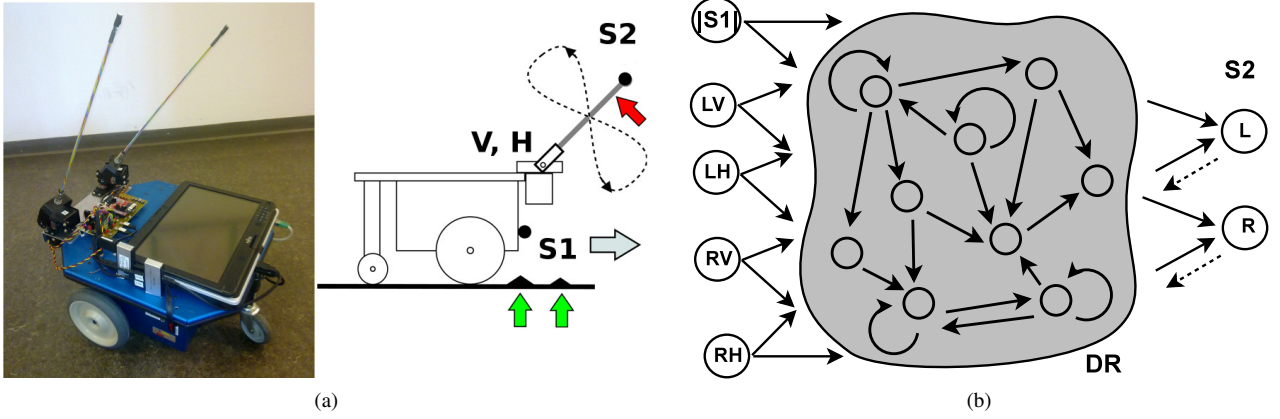


Figure 1: (a): Experimental set-up showing the ERA robot and the positions of the sensors. S1 and S2 are the acceleration sensors mounted to the robot platform and to the tip of the antenna, respectively. Arrows correspond to possible causes of events, touch (red) and disturbances (green), recorded by S2 sensor signal. (b): Echo State Network showing the inputs, S1 and antennal servo commands (LV, LH, RV, RH), Dynamical reservoir (DR), and the outputs S2. L and R represent left and right antennae. V and H suffixes stand for vertical and horizontal.

accelerometer (ADXL321) mounted at the tip of a 40 cm poly-acrylic tube. Both antennae were moving in an “8”-shaped pattern with an approximate frequency of 0.56 Hz. The working-range of each antenna was about 100 degrees in both the vertical and horizontal planes. An Inertial Measurement Unit (x-IMU, from *x-io Technologies Ltd.*, which includes a 3-axes acceleration sensor) was mounted on the robot (S1) in order to measure the vibration of the robot base while moving. The readout for the acceleration sensors was implemented on a micro-controller (ATMEL AT90CAN128) and the raw sensor signals were transferred to a notebook via serial interface (RS232/USB) for storage and further processing in Matlab (7.9.0-R2009b). All the devices were integrated into a common framework for easy communication.

The servo position commands and both acceleration data were preprocessed in order to remove the DC-bias. Furthermore, the two components of S2 (X, Y) and the components of S1 (X, Y, Z) were combined according to the *Euclidean norm* before being fed into the ESN model. As shown in Figure 1b, a single network was used to predict the accelerometer readings of both antennae.

The Matlab ESN toolbox created by D. Popovici (revised by H. Jaeger) was used to implement the ESN [7]. The parameters, such as size of the dynamic reservoir (DR), spectral radius (SR) and input scaling factor (ISF) were systematically varied in order to find a suitable set for the ESN. The normalized root mean square error (NRMSE) between the predicted and actual time series was used as the performance measure. Then, the trained network was used as the forward model in the proposed method for detecting true touch events to the antennal system (see Figure 2 and section 3).

2.1 Experiments

The forward-model-based prediction system was tested in three different scenarios:

- **Straight motion** - The robot was moving straight on a flat surface with a constant velocity of 0.3 ms^{-1} . Obstacles or bumps (5 mm height) were placed along the robot’s pathway to generate disturbances. A wooden stick was used as the contact object presented to the antennae.
- **Curve motion** - In separate trials, the robot was set to move on a curved path in clockwise or in anti-clockwise direction. Here, the linear and angular velocities were about 0.15 ms^{-1} and 0.5 rads^{-1} (approximately 30 degrees/s) respectively. The bumps and contact objects were introduced as in the previous case.
- **Spiked wheels**: A small rubber protrusion (about 1 cm x 1 cm x thickness of the wheel) was attached to each of the front wheels of the ERA robot with a phase shift of 180 degrees. When the robot moved forward, the wheels alternatingly generated rhythmic impacts. With this experiment, we were able to estimate the performance of an active tactile system mounted to a walking robot with rhythmic impacts, as caused by the limbs touching the ground. Here, the robot was moving straight with a constant velocity of approximately 0.2 ms^{-1} .

3 Results and Discussion

When the wheeled robot drove across bumps, the impact caused vibrations, thus affecting the tactile sensor reading (S2). We refer to these unwanted sensory readings as

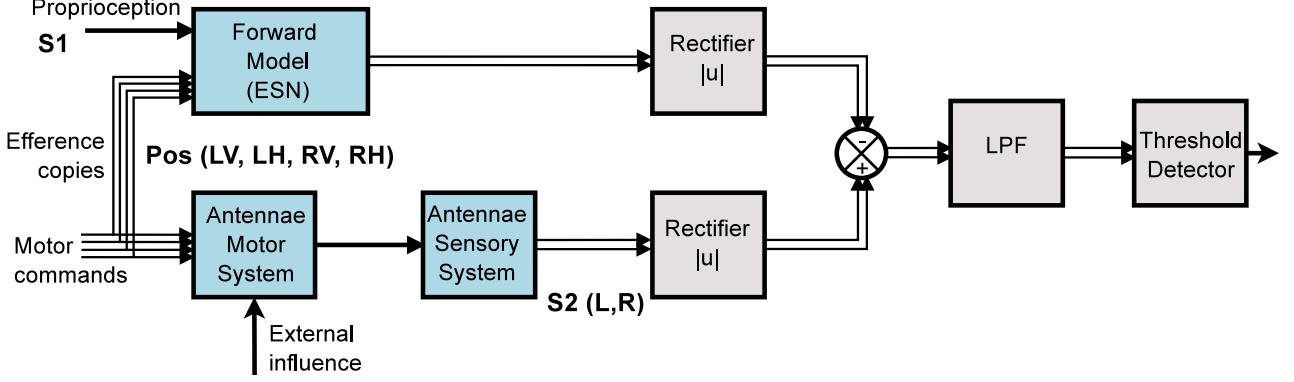


Figure 2: Our approach for the detection of true touch events to the antennae. An ESN-based forward model is used to predict the expected sensory reading of S2 due to ego motion. Rectifiers are used for obtaining absolute values of the signals. S1, S2 and the position or motor commands, LV, LH, RV and RH have the same meanings as in Figure 1b. LPF: Low Pass Filter.

disturbances. On the other hand, the antennal sensors (S2) were capturing direct contacts (touch events) to the antennae. Owing to the similar magnitude of these two types of responses, we could not directly use a threshold on the S2 signal since that would have caused miss-classification of disturbances as touch events. Therefore, we applied the forward model for predicting the expected S2 sensory reading and then the reading was compared with the actual value for peak cancellation. A small but variable phase difference could be seen between the *actual* and the *predicted* signals. Therefore, simple subtraction of the predicted signal from the actual traces did not cancel all unwanted peaks in the original signal. As shown in Figure 2, both signals were first *rectified*, then subtracted and finally filtered with a first-order low-pass filter (LPF). In fact, for a linear filter, this is same as taking the envelopes before the subtraction. We tested the model performance for LPFs with different cut-off frequencies (ω_c): 100, 50, 20 and 10 Hz. Table 1 summarizes the number of total false predictions in the three scenarios used, i.e., *straight* and *curved* trials with bumps on the floor, and straight trials with *spiked* wheels (see section 2.1). With cut-off frequencies 100 and 50 Hz, the prediction system was able to correctly identify the true touch events, i.e., without false negatives. However,

some of the the disturbances were misclassified as touch events (false positives). In summary, we found that a cut-off frequency of 20 Hz allowed the forward model to distinguish 53 touch events from 252 other disturbances with a success rate of about 96%. Note that the touch events occurred at different positions along the each antenna covering approximately 20-90% (proximal to distal) of its length. For a lower ω_c , e.g., $\omega_c = 10$ Hz, false negatives occur owing to stronger smoothing of the input signal. Given the high performance for single contact events, we conclude that our method provides an appropriate means of eliminating self-stimulation of a tactile sensor on a moving body. Its neural network implementation makes the forward model a good candidate for prediction and elimination of self-induced stimulation of mechanosensors in general, but also for sensors in other modalities.

Our approach implements a forward model using efference copies (LV, LH, RV, RH) and a local proprioceptive signal (S1) as inputs. It is interesting to compare the importance of each signal as an input to the model. Therefore, we conducted simulated *lesion experiments* and compared how the individual inputs affect the prediction performance. In each trial, at first, the ESN was trained using both signals as inputs. Then the prediction task was carried out with the model receiving only one input, i.e., either the local proprioception or the efference copies (all four) were set to zero or to a constant value. Table 2 lists the number of false predictions that occurred in each scenario. Two cases are compared; one, where the efference copies were set to 0.5 and one where they were set to zero. Thus, the only varying input available to the ESN was the local proprioception from S1. In both cases, the lowest number of misclassifications increased compared to the full model, and the prediction performance declined by about 7.5 or 5.0%, depending on whether the ECs were non-zero or zero. The minimum occurred for $\omega_c = 10$ or 20 Hz respectively. On the other hand, without the S1 signal (after setting it to either 0.0 or 0.5), the ESN was completely unable to predict

No.Trials		No. of Events		ω_c [Hz]			
		Touch	Disturbances	100	50	20	10
Straight n = 16	L	16	44	26	14	3	11
	R	11	44	27	14	0	6
Curve n = 10	L	13	35	18	11	4	9
	R	13	35	17	10	1	2
Spiked n = 3	L	0	47	28	22	1	2
	R	0	47	31	14	3	1
Total		53	252	147	85	12	31

Table 1: Total number of false predictions in each scenario

No.Trials		ω_c [Hz]							
		100		50		20		10	
		0.5	0	0.5	0	0.5	0	0.5	0
Straight n = 16	L	35	25	23	16	12	8	10	11
	R	36	31	26	16	9	4	7	6
Curve n = 10	L	23	16	18	11	8	4	8	7
	R	22	14	14	8	6	2	6	3
Spiked n = 3	L	24	23	19	18	8	6	3	4
	R	22	19	18	18	4	2	1	1
Total		162	128	118	87	47	26	35	32

Table 2: False predictions of the model with efference copies (normalized EC) set to either 0.5 or 0.0

the effect on the S2 sensors due to disturbances. Hence, the proposed method (see Figure 2) failed to distinguish touch events to the antennae from other disturbances. Therefore we conclude that, in the current experimental setup, both the local proprioceptive signal and the efference copies are useful but only the proprioceptive signal is necessary. Nevertheless, the efference copy is a *necessary* input to the ESN in order to predict the induced accelerations due to self-motion (oscillatory movements when sampling the 3D space) of the active tactile system.

4 Summary

In this study, we expand our work on Echo State Network (ESN) based forward models for predicting the expected sensor reading of an insect-inspired active tactile sensor system on a mobile robot [3]. ESNs are a special kind of recurrent artificial neural networks that are suitable for chaotic time series prediction. Inputs to the forward model are the motor commands which set the positions of both antennae, and a local proprioceptive signal which measures the vibrations of the robot platform. The model was successfully used to detect tactile events on each antenna while the robot, is either moved on a straight or curved path with bumps on the floor, or moved with spiked wheels. The latter scenario was chosen to estimate the performance of an active tactile system mounted to a walking robot with rhythmic impacts caused by the legs touching down. By simulated lesion experiments, we found that the local proprioceptive signal plays a major role in discriminating disturbances from tactile events in the proposed forward model. We conclude that such an ESN-based forward model is suitable for prediction and elimination of self-induced stimulation of sensors, in particular of mechanosensors.

5 Open Questions

In this study, we have tested the model on real measured data that were analyzed offline. Online implementation will require real-time control and computational efficient libraries for ESN etc. Also, putting controllers and the forward model

into a single embedded system would be advantageous. Another point to discuss is the selection of the cut-off frequency (ω_c), which has a crucial role for the overall performance, of the LPF. It is known that the decay time of the antennal oscillations depend on the contact position along the antenna [2]. Therefore, the use of an adaptive ω_c as opposed to a globally fixed value could be advantageous. Furthermore, we know that the signal to be predicted (S2) has both fast and slow dynamics. In general, the spectral radius (SR) of the ESN should be small for the fast dynamics and large for the slow dynamics. Hence, finding a single SR value that is equally well suited for different frequency components is difficult. Recently, [8] introduced a modular ESN architecture comprising a combination of local expert ESNs with different SR values and one special gating ESN. Such a method will also serve to improve the performance of the forward model discussed here. Moreover, the adaptation of the method to other sensory modalities, such as vision, may be considered.

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